

1 Software Requirements Specification: FindBack

System: FindBack — Campus Lost & Found

Version: 1.0

Course: CSCD602: Advanced Software Engineering

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1.1 1. Purpose and Scope

FindBack provides a trusted campus workflow for reporting lost or found property, discovering relevant reports, proving ownership, approving or rejecting claims, and protecting reporter contact details. Its users are students and administrators. The system is intentionally limited to a 48-hour capstone scope: a responsive web interface, managed sign-in, report and claim workflows, optional single-image attachment, in-app notifications and administrator control.

1.2 2. Users and Roles

Role	Capabilities
Public visitor	Browse, search and filter reports; view non-sensitive item details.
Student	Use all public features; create reports; submit claims; view personal reports, claims and notifications.
Admin	View protected reporter contact data; review claims; update item statuses; manage report lifecycle.

1.3 3. Functional Requirements

ID	Requirement	Priority
FR-01	The system shall authenticate users and enforce student and admin roles at the server.	Must
FR-02	A student shall create a lost-item report with title, description, category, date, location and optional permitted image.	Must
FR-03	A student shall create a found-item report with the same required fields.	Must
FR-04	The system shall store an optional PNG, JPEG or WebP	Must

ID	Requirement	Priority
	image no larger than 4 MB in managed object storage.	
FR-05	Public visitors shall browse reports by keyword, category, status and date using paginated results.	Must
FR-06	An authenticated student shall submit a single claim with ownership evidence for an active found report they did not create.	Must
FR-07	An admin shall approve or reject a pending claim and may provide a decision note.	Must
FR-08	An admin shall update a report only to lost, found, resolved or archived.	Must
FR-09	Reporter contact information shall remain hidden unless the requester is an admin or holds an approved claim for the exact item.	Must
FR-10	The system shall create an in-app notification when a claim is approved or rejected.	Must
FR-11	A student shall view own reports, claims, decision notes and notifications on a profile page.	Must
FR-12	Mutating actions shall provide clear success or error feedback through toast notifications.	Should

1.4 4. Non-Functional Requirements

ID	Requirement	Acceptance measure
NFR-01	Responsive experience	Important pages remain usable at mobile and desktop widths.

ID	Requirement	Acceptance measure
NFR-02	Security	Role checks and contact-data policy are enforced in server procedures.
NFR-03	Input integrity	Schemas validate required text, dates, status values, ownership proof and image data.
NFR-04	Privacy	Public results never include reporter e-mail/contact data.
NFR-05	Usability	Forms use labels, actionable errors, loading states, empty states and visible feedback.
NFR-06	Maintainability	Typed contracts, small domain rules, automated tests and documented technical debt support future maintenance.

1.5 5. Data Requirements

The core data entities are Users, Items, Claims and Notifications. An Item belongs to a reporting User. A Claim belongs to a claimant User and an Item; the (itemId, claimantId) pair is unique to prevent duplicate claims. A Notification belongs to a User and references the decision-related Item and Claim. Actual image bytes are stored outside the relational database, while only image identifiers and URLs are persisted.

1.6 6. Constraints and Assumptions

The system relies on the managed platform's OAuth session, relational database and object storage. A user can have either student or admin role. In-app notifications are delivered when the recipient next opens the application; the system does not send e-mail, SMS or push notifications in version 1.0. The item directory is intentionally simplified for the expected examination-scale data volume.

1.7 7. Acceptance Summary

The application is acceptable when authenticated report creation, public discovery, claim restriction, admin decision, in-app notification, privacy-controlled contact revelation and permitted-status validation all operate on the published URL. The final live acceptance tests are recorded in `Testing_Report.pdf`.